$$\begin{aligned}\sum_{n=1}^{\infty} P(X \geq n) &= \sum_{n=1}^{\infty}\sum_{r=n}^{\infty} P(X=r) \\ &= \sum_{r=1}^{\infty}\sum_{n=1}^{r} P(X=r) \\ &= \sum_{r=1}^{\infty} rP(X=r) \\ &= \mathrm{E}(X)\end{aligned}$$

$X \geq 4 \iff$ first three boxes are all daddy or all mummy

$$P(X \geq 4) = p^3 + q^3$$

$X \geq n \iff$ first $n-1$ boxes are all daddy or all mummy

$$P(X \geq n) = p^{n-1} + q^{n-1}$$

$$P(X \geq 1) = 1,\ P(X \geq 2) = 1$$

$$\begin{aligned}\mathrm{E}(X) &= 1 + 1 + \sum_{n=3}^{\infty}\left(p^{n-1} + q^{n-1}\right) \\ &= 2 + \frac{p^2}{1-p} + \frac{q^2}{1-q} \\ &= 2 + \frac{p^2}{q} + \frac{q^2}{p} \\ &= 2 + \frac{p^3 + q^3}{pq}\end{aligned}$$

$$p + q = 1 \implies p^3 + q^3 = (p+q)^3 - 3pq(p+q) = 1 - 3pq$$

$$\therefore \mathrm{E}(X) = 2 + \frac{1-3pq}{pq} = \frac{1}{pq} - 1$$

$$pq \leq \left(\frac{p+q}{2}\right)^2 = \frac{1}{4}$$

$$\therefore \mathrm{E}(X) = \frac{1}{pq} - 1 \geq 4 - 1 = 3$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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